Ahmed Iprahim Abdelhamed

Junior Data Scientist

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SUMMARY

Results-driven Data Scientist with a B.Sc. in Computer Science and ALX certification in Machine Learning and Data Science. Experienced in building end-to-end ML models including hybrid recommender systems, regression pipelines, classification algorithms, and predictive analytics solutions. Strong foundation in Python, Scikit-learn, XGBoost, SQL, and statistical modeling, with hands-on experience applying data-driven approaches to fintech operations at Tabby.

SKILLS

Machine Learning & Modeling: Supervised/Unsupervised Learning, Feature Engineering, XGBoost, Regression/Classification, SVD, Hyperparameter Tuning
Languages & Querying: Python (Pandas, NumPy, Scikit-Learn, Statsmodels), SQL (PostgreSQL, DuckDB), Bash, Git
Visualization & Deployment: Streamlit, Power BI, Looker Studio, Plotly, Matplotlib, Seaborn
Data Engineering & Diagnostics: ETL Pipelines, EDA, VIF Analysis, LINE Framework, Statistical Testing

PROJECTS

Credit Risk & Expected Loss AI Engine | *Python, XGBoost, PostgreSQL, Streamlit, Scikit-learn*

- Engineered an end-to-end classification pipeline predicting applicant Probability of Default (PD) on 32,000+ financial records while preventing data leakage by dropping post-outcome metrics.
- Handled severe class imbalance (78% good / 22% default) using XGBoost scale_pos_weight and GridSearchCV, boosting default recall to 78% without sacrificing model precision.
- Deployed the trained model as an interactive Streamlit web application, enabling real-time credit scoring and automated risk classification for new loan applicants.

CineMatch AI - Hybrid Movie Recommender System | *Python, Scikit-learn, Pandas, NumPy, Streamlit*

- Built an end-to-end hybrid recommender processing 10M+ MovieLens ratings using SVD (bias + latent factors).
- Achieved an RMSE of 0.8794, outperforming baseline models by 17% through feature engineering and hyperparameter tuning.
- Deployed an interactive Streamlit web application featuring fuzzy string search and real-time similarity matching.

Ames Housing Valuation & Regression Diagnostics | *Python, Scikit-learn, Statsmodels, Streamlit*

- Built an end-to-end regression pipeline predicting real estate valuations while optimizing model complexity and signal density.
- Reduced feature space from 37 to 24 by diagnosing correlation matrices and Variance Inflation Factors (VIF) to eliminate 13 multicollinear variables.
- Validated LINE regression assumptions and engineered 4 interaction/polynomial features, boosting R^2 from 0.8356 to 0.8999.
- Deployed an interactive Streamlit application enabling real-time valuation estimates for stakeholders.

PROFESSIONAL EXPERIENCE

Tabby | Data & Analytics Specialist | *01/2024 - Present*

- Built Python scripts to extract, clean, and aggregate multi-channel social engagement datasets of over 50,000+ records for quantitative performance analysis.
- Automated weekly reporting pipelines using Python and SQL, eliminating manual data collection and reducing reporting time by 10+ hours per week.
- Performed time-series and statistical trend analysis on 10,000+ weekly customer inquiry volumes, accurately forecasting demand spikes and optimizing operational capacity.

Tabby | Social Media & Customer Support Specialist | *01/2023 - 12/2023*

- Analyzed historical performance metrics across chat, email, and phone channels to identify service level variance, consistently maintaining 92%+ CSAT.
- Managed real-time interaction datasets and pattern anomalies to maintain SLA adherence during volume spikes.

EDUCATION & CERTIFICATIONS

Helwan University | Bachelor of Science in Computer Science | *2024* | Helwan, Egypt
Certifications: Data Science Certificate (ALX - 2024) | Machine Learning Certificate (ALX - 2024) | Data Analytics Professional Certificate (ALX - 2024) | Python Programming Certificate (ALX - 2024)